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(54) **COMPOSITE ELECTRODE**

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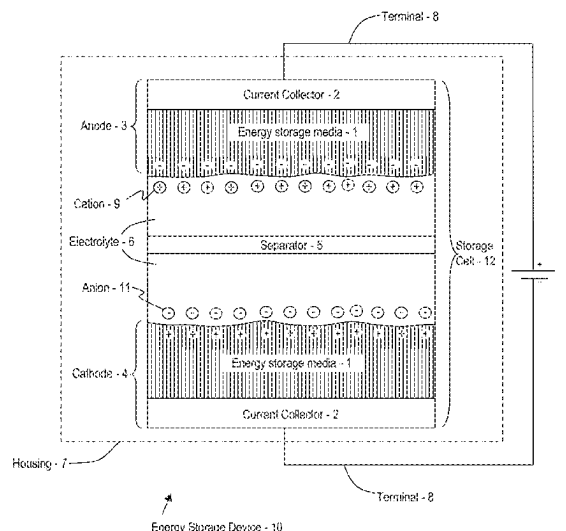
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(57) **ABSTRACT**

An apparatus is disclosed that includes an active storage layer including: a network of carbon nanotubes defining void spaces; and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes. In some cases, the active layer provides energy storage, e.g., in an ultracapacitor device.

12 Claims, 6 Drawing Sheets



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Fig. 1

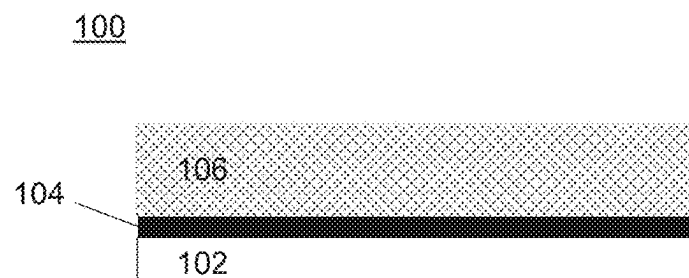


Fig. 2

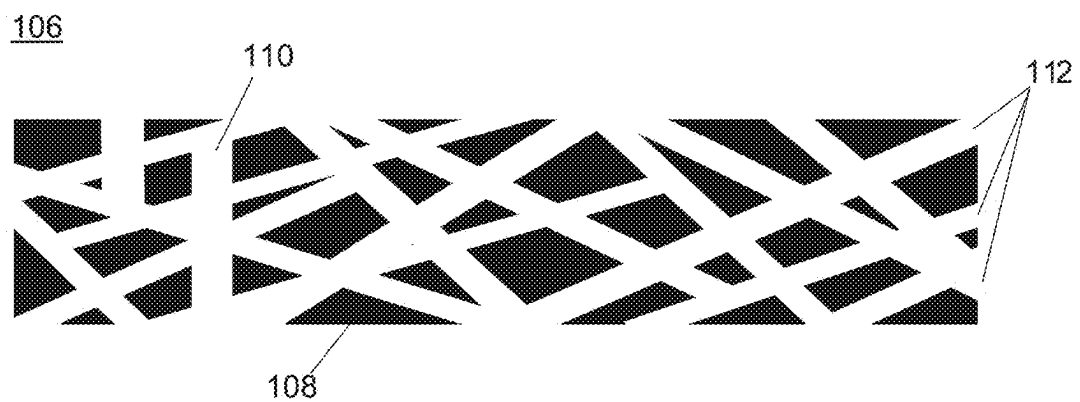


Fig. 3

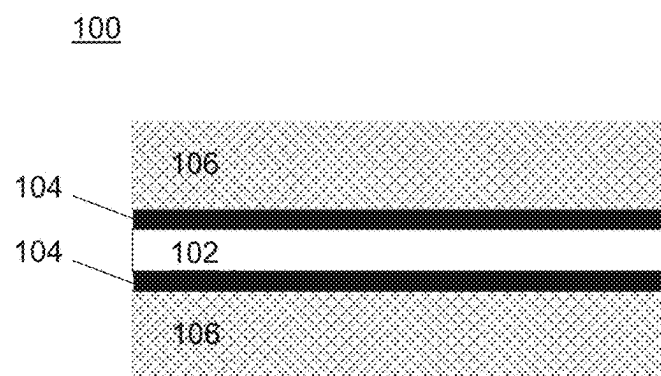


Fig. 4

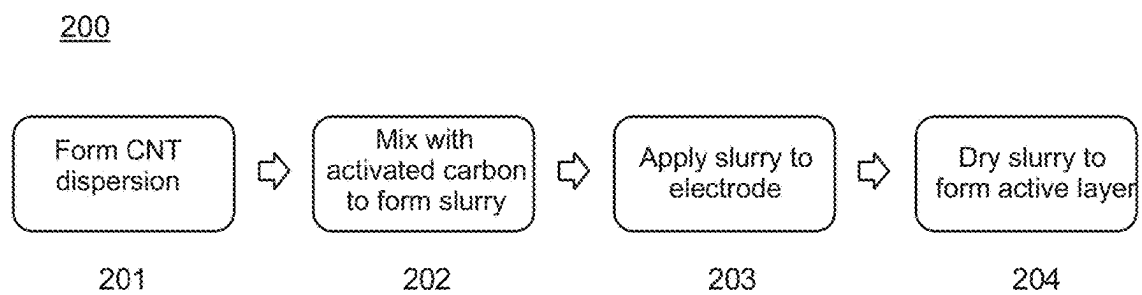


Fig. 5

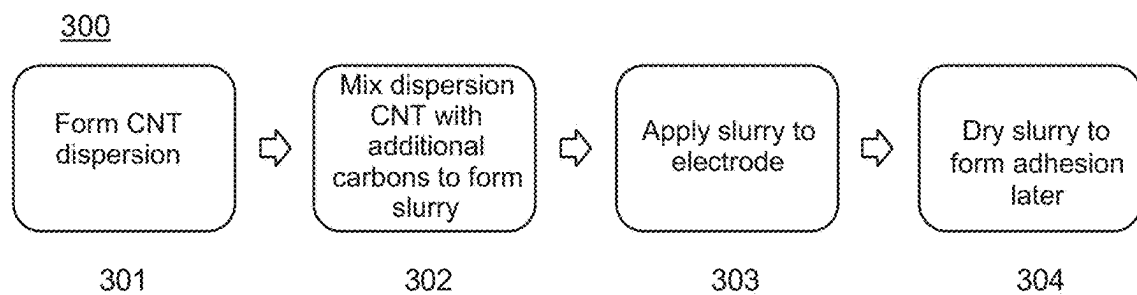


Fig. 6

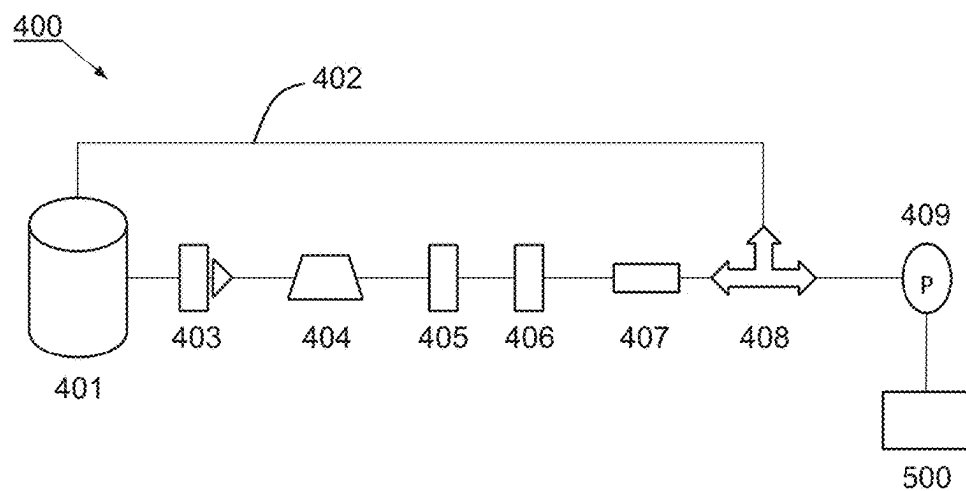


Fig. 7A 500

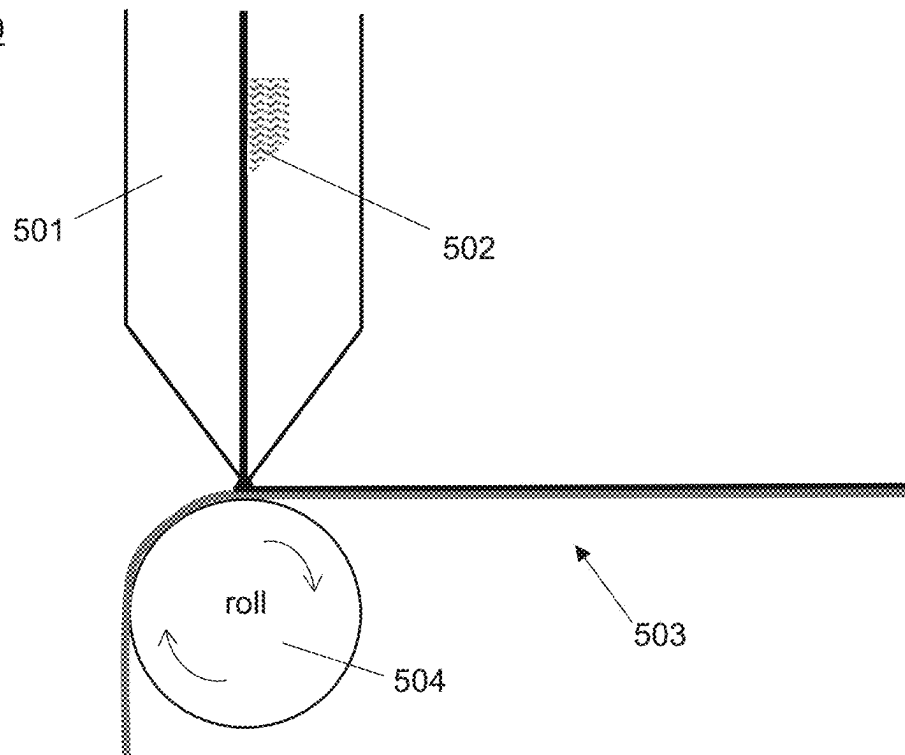


Fig. 7B

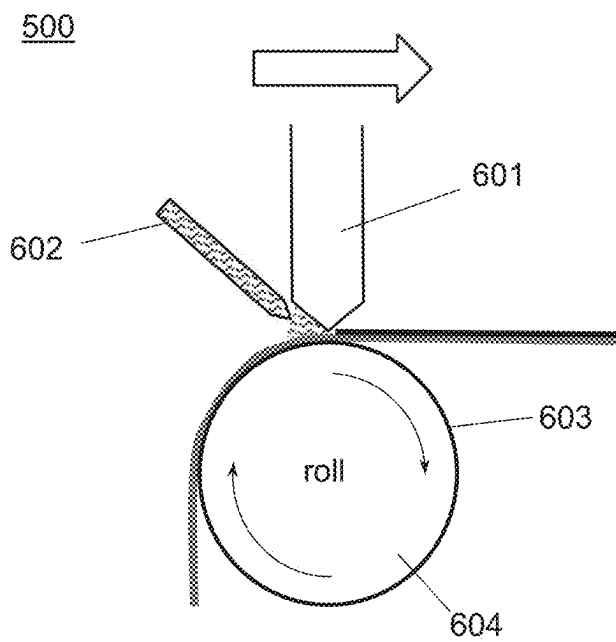


Fig. 8A

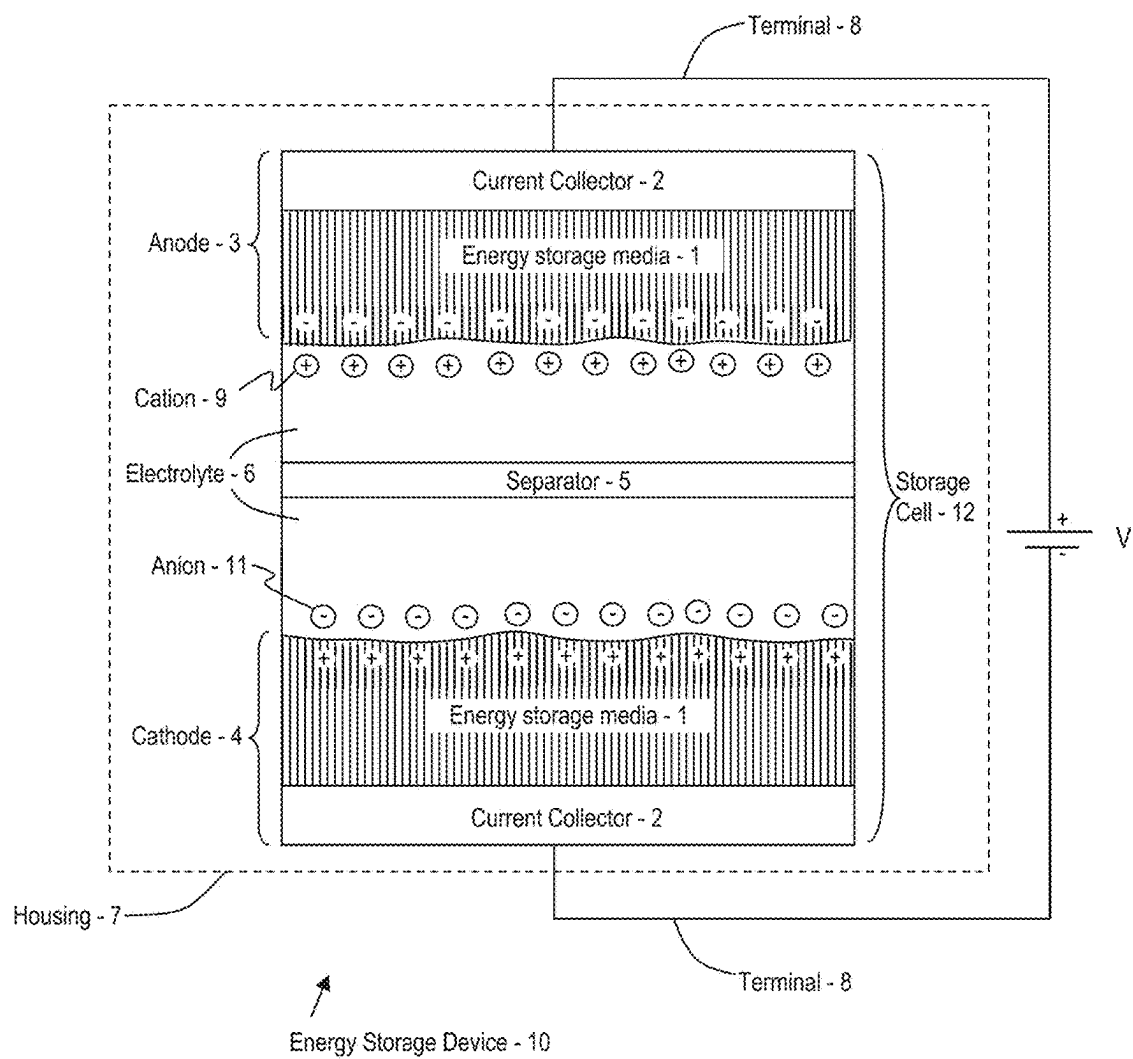
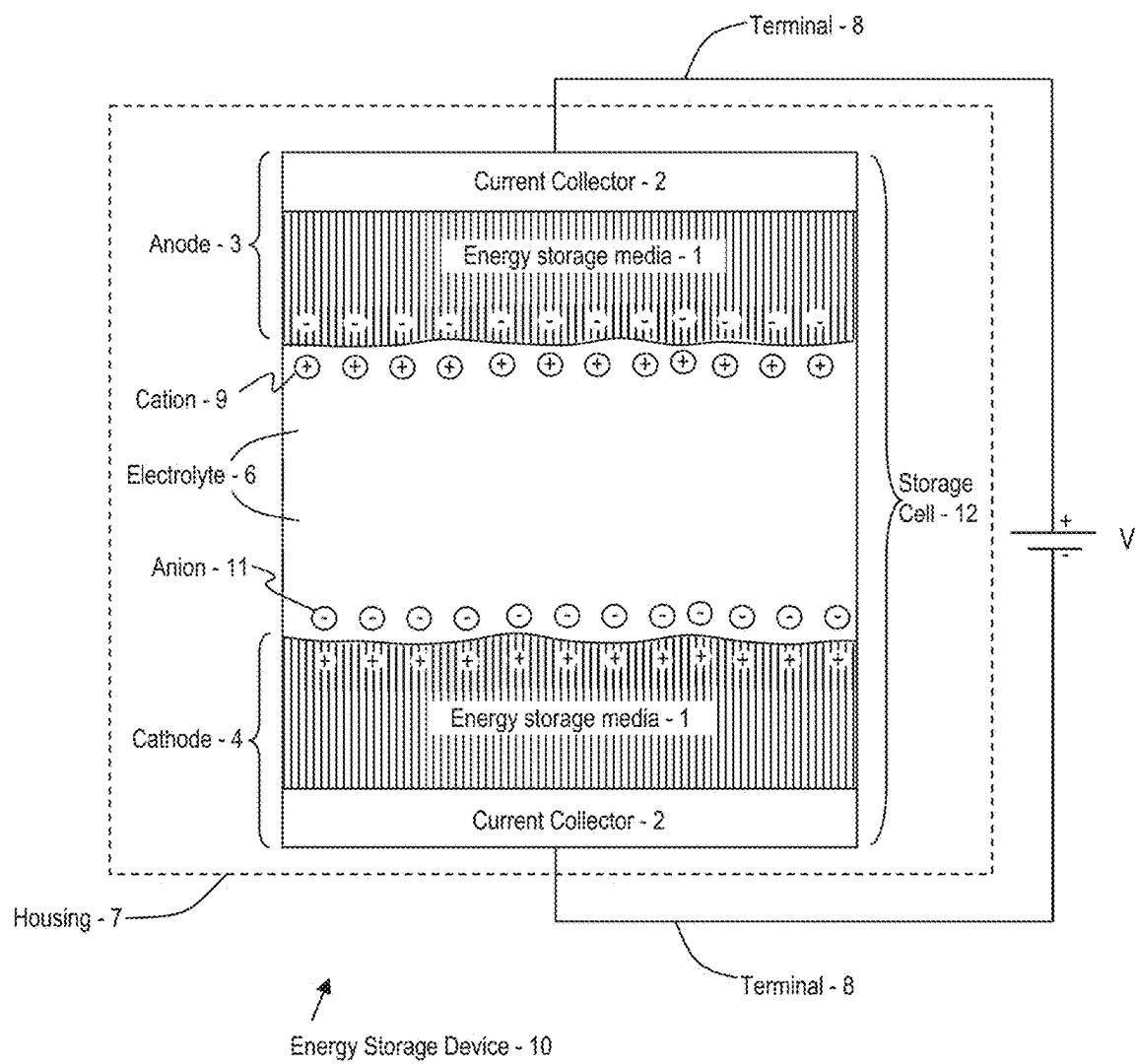


Fig. 8B



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COMPOSITE ELECTRODE**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation of U.S. patent application Ser. No. 16/681,293, filed Nov. 12, 2019, now U.S. Pat. No. 10,600,582, granted Mar. 24, 2020, which is a continuation of U.S. patent application Ser. No. 16/427,546 filed on May 31, 2019, which is a continuation-in-part of PCT/US2017/064152 filed on Dec. 1, 2017, which claims the benefit of U.S. Provisional Application No. 62/429,727, filed Dec. 2, 2016, the entire contents of which are incorporated herein by reference.

BACKGROUND

Carbon nanotubes (hereinafter referred to also as "CNTs") are carbon structures that exhibit a variety of properties. Many of the properties suggest opportunities for improvements in a variety of technology areas. These technology areas include electronic device materials, optical materials as well as conducting and other materials. For example, CNTs are proving to be useful for energy storage in capacitors.

However, CNTs are typically expensive to produce and may present special challenges during electrode manufacturing. Accordingly, there is a need for an electrode material that exhibits the advantageous properties of CNTs while mitigating the amount of CNTs included in the material.

SUMMARY

The applicants have developed a composite electrode structure that exhibits advantageous properties. In some embodiments, the electrode exhibits the advantageous properties of CNTs while mitigating the amount of CNTs included in the material, e.g., to less than 10% by weight.

Electrodes of the type described herein may be used in ultracapacitors to provide high performance (e.g., high operating, voltage, high operating temperature, high energy density, high power density, low equivalent series resistance, etc.).

In one aspect, an apparatus is disclosed including an active storage layer including a network of carbon nanotubes defining void spaces; and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes, wherein the active layer is configured to provide energy storage.

In some embodiments, the active layer is substantially free from binding agents. In some embodiments, the active layer consists of or consists essentially of carbonaceous material. In some embodiments, the active layer is bound together by electrostatic forces between the carbon nanotubes and the carbonaceous material. In some embodiments, the carbonaceous material includes activated carbon.

In some embodiments, the carbonaceous material includes nanoform carbon other than carbon nanotubes.

In some embodiments, the network of carbon nanotubes makes up less than 50% by weight of the active layer, less than 10% by weight of the active layer, less than 5% by weight of the active layer, or less than 1% by weight of the active layer.

Some embodiments include an adhesion layer, e.g., a layer consisting of or consisting essentially of carbon nanotubes. In some embodiments the adhesion layer is disposed between the active layer and an electrically conductive layer.

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In some embodiments, a surface of the conductive layer facing the adhesion layer includes a roughened or textured portion. In some embodiments, a surface of the conductive layer facing the adhesion layer includes a nanostructured portion. In some embodiments, the nanostructured portion includes carbide "nanowhiskers". These nanowhiskers are thin elongated structures (e.g., nanorods) that extend generally away from the surface of the conductor layer 102. The nanowhiskers may have a radial thickness of less than 100 nm, 50 nm, 25, nm, 10 nm, or less, e.g., in the range of 1 nm to 100 nm or any subrange thereof. The nanowhisker may have a longitudinal length that is several to many times its radial thickness, e.g., greater than 20 nm, 50 nm, 100 nm, 200 nm, 300, nm, 400, nm, 500 nm, 1 μ m, 5 μ m, 10 μ m, or more, e.g., in the range of 20 nm to 100 μ m or any subrange thereof.

In some embodiments, the active layer has been annealed to reduce the presence of impurities.

In some embodiments, active layer has been compressed to deform at least a portion of the network of carbon nanotubes and carbonaceous material.

Some embodiments include an electrode including the active layer. Some embodiments include an ultracapacitor including the electrode. In some embodiments, the ultracapacitor has an operating voltage greater than 1.0 V, 2.0 V, 2.5 V, 3.0 V, 3.1 V, 3.2 V, 3.5 V, 4.0 V or more.

In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 1.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 2.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 3.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 4.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 1.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 2.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 3.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 4.0 V for a lifetime of at least 1,000 hours.

In another aspect, a method including: dispersing carbon nanotubes in a solvent to form a dispersion; mixing the dispersion with carbonaceous material to form a slurry; applying the slurring in a layer; and drying the slurry to substantially remove the solvent to form an active layer including a network of carbon nanotubes defining void spaces and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes. Some embodiment include forming or applying a layer of carbon nanotubes to provide an adhesion layer on a conductive layer.

In some embodiments, the applying step including applying the slurry onto the adhesion layer.

Various embodiments may include any of the forgoing elements or features, or any elements or features described herein either alone or in any suitable combination.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic of an electrode.

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FIG. 2 is an illustration of a detailed view of an active layer of an electrode.

FIG. 3 is a schematic of a two-sided electrode.

FIG. 4 is a flow chart illustrating a method of making an active layer for an electrode.

FIG. 5 is a flow chart illustrating a method of making an adhesion layer for an electrode.

FIG. 6 is a schematic diagram of an exemplary mixing apparatus.

FIG. 7A is a schematic diagram of coating apparatus featuring a slot die.

FIG. 7B is a schematic diagram of coating apparatus featuring a doctor blade.

FIG. 8A is a schematic of an ultracapacitor.

FIG. 8B is a schematic of an ultracapacitor without a separator.

DETAILED DESCRIPTION

Referring to FIG. 1, an exemplary embodiment of an electrode 100 is disclosed for use in an energy storage device, such as an ultracapacitor or battery. The electrode includes an electrically conductive layer 102 (also referred to herein as a current collector), an adhesion layer 104, and an active layer 106. When used in an ultracapacitor of the type described herein, the active layer 106 may act as energy storage media, for example, by providing a surface interface with an electrolyte (not shown) for formation of an electric double layer (sometimes referred to in the art as a Helmholtz layer). In some embodiments, the adhesion layer 104 may be omitted, e.g., in cases where the active layer 106 exhibits good adhesion to the electrically conductive layer 102.

In some embodiments, the active layer 106 may be thicker than the adhesion layer 104, e.g., 1.5, 2.0, 5.0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1,000 or more times the thickness of the adhesion layer 104. For example, in some embodiments, the thickness of the active layer 106 may be in the range of 1.5 to 1,000 times the thickness of the adhesion layer 104 (or any subrange thereof, such as 5 to 100 times). For example, in some embodiments the active layer 106 may have a thickness in the in the range of 0.5 to 2500 μm or any subrange thereof, e.g., 5 μm to 150 μm . In some embodiments the adhesion layer 104 may have a thickness in the range of 0.5 μm to 50 μm or any subrange thereof, e.g., 1 μm to 5 μm .

Referring to FIG. 2, in some embodiments, the active layer 106 is comprised of carbonaceous material 108 (e.g., activated carbon) bound together by a matrix 110 of CNTs 112 (e.g., a webbing or network formed of CNTs). In some embodiments, e.g., where the length of the CNTs is longer than the thickness of the active layer 106, the CNTs 112 forming the matrix 110 may lie primarily parallel to a major surface of the active layer 106. Not that although as shown the CNTs 112 form straight segments, in some embodiments, e.g., where longer CNTs are used, the some or all of the CNTs may instead have a curved or serpentine shape. For example, in cases where the carbonaceous material 108 includes lumps of activated carbon, the CNTs 112 may curve and wind between the lumps.

In some embodiments, the active layer is substantially free of any other binder material, such as polymer materials, adhesives, or the like. In other words, in such embodiments, the active layer is substantially free from any material other than carbon. For example, in some embodiments, the active layer may be at least about 90 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt %, 99.5 wt %, 99.9 wt %, 99.99 wt %, 99.999 wt %, or more elemental carbon by mass. Despite

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this, the matrix 110 operated to bind together the carbonaceous material 108, e.g., to maintain the structural integrity of the active layer 106 without flaking, delamination, disintegration, or the like.

It has been found that use of an active layer substantially free of any non-carbon impurities substantially increases the performance of the active layer in the presence of high voltage differentials, high temperatures, or both. Not wishing to be bound by theory, it is believed that the lack of impurities prevents the occurrence of unwanted chemical side reactions which otherwise would be promoted in high temperature or high voltage conditions.

As noted above, in some embodiments, the matrix 110 of carbon nanotubes provides a structural framework for the active layer 106, with the carbonaceous material 108 filling the spaces between the CNTs 112 of the matrix 110. In some embodiments, electrostatic forces (e.g., Van Der Waals forces) between the CNTs 112 within the matrix 110 and between the matrix 112 and the other carbonaceous material 108 may provide substantially all of the binding forces maintaining the structural integrity of the layer.

In some embodiments, the CNTs 112 may include single wall nanotubes (SWNT), double wall nanotubes (DWNT), or multiwall nanotubes (MWNT), or mixtures thereof. Although a matrix 110 of individual CNTs 112 is shown, in some embodiments, the matrix may include interconnected bundles, clusters or aggregates of CNTs. For example, in some embodiments where the CNTs are initially formed as vertically aligned, the matrix may be made up at least in part of brush like bundles of aligned CNTs.

In order to provide some context for the teachings herein, reference is first made to U.S. Pat. No. 7,897,209, entitled "Apparatus and Method for Producing Aligned Carbon Nanotube Aggregate." The foregoing patent (the "'209 patent") teaches a process for producing aligned carbon nanotube aggregate. Accordingly, the teachings of the '209 patent, which are but one example of techniques for producing CNTs in the form of an aligned carbon nanotube aggregate, may be used to harvest CNTs referred to herein. Advantageously, the teachings of the '209 patent may be used to obtain long CNTs having high purity. In other embodiments, any other suitable method known in the art for producing CNTs may be used.

In some embodiments the active layer 106 may be formed as follows. A first solution (also referred to herein as a slurry) is provided that includes a solvent and a dispersion of carbon nanotubes, e.g., vertically aligned carbon nanotubes. A second solution (also referred to herein as a slurry) may be provided that includes a solvent with carbon disposed therein. This carbon addition includes at least one form of material that is substantially composed of carbon. Exemplary forms of the carbon addition include, for example, at least one of activated carbon, carbon powder, carbon fibers, rayon, graphene, aerogel, nanohorns, carbon nanotubes and the like. While in some embodiments, the carbon addition is formed substantially of carbon, it is recognized that in alternative embodiments the carbon addition may include at least some impurities, e.g., additives included by design.

In some embodiments, forming the first and/or second solution include introducing mechanical energy into the mixture of solvent and carbon material, e.g., using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy introduced into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy

introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

In some embodiments, the solvents used may include an anhydrous solvent. For example, the solvent may include at least one of ethanol, methanol, isopropyl alcohol, dimethyl sulfoxide, dimethylformamide, acetone, acetonitrile, and the like.

As noted above, the two solutions may be subjected to “sonication” (physical effects realized in an ultrasonic field). With regard to the first solution, the sonication is generally conducted for a period that is adequate to tease out, fluff or otherwise parse the carbon nanotubes. With regard to the second solution, the sonication is generally conducted for a period that is adequate to ensure good dispersion or mixing of the carbon additions within the solvent. In some embodiments, other techniques for imparting mechanical energy to the mixtures may be used in addition or alternative to sonication, e.g., physical mixing using stirring or impeller.

Once one or both of the first solution and the second solution have been adequately sonicated, they are then mixed together, to provide a combined solution and may again be sonicated. Generally, the combined mixture is sonicated for a period that is adequate to ensure good mixing of the carbon nanotubes with the carbon addition. This second mixing (followed by suitable application and drying steps as described below) results in the formation of the active layer 106 containing the matrix 110 of CNTs 112, with the carbon addition providing the other carbonaceous material 108 filling the void spaces of the matrix 110.

In some embodiments, mechanical energy may be introduced to the combined mixture using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

In some embodiments, the combined slurry may be cast wet directly onto the adhesion layer 104 or the conductive layer 102, and dried (e.g., by applying heat or vacuum or both) until substantially all of the solvent and any other liquids have been removed, thereby forming the active layer 106. In some such embodiments it may be desirable to protect various parts of the underlying layers (e.g., an underside of a conductive layer 102 where the current collector is intended for two sided operation) from the solvent, e.g., by masking certain areas, or providing a drain to direct the solvent.

In other embodiments, the combined slurry may be dried elsewhere and then transferred onto the adhesion layer 104 or the conductive layer 102 to form the active layer 106, using any suitable technique (e.g., roll-to-roll layer application). In some embodiments the wet combined slurry may be placed onto an appropriate surface and dried to form the active layer 106. While any material deemed appropriate may be used for the surface, exemplary material includes PTFE as subsequent removal from the surface is facilitated by the properties thereof. In some embodiments, the active layer 106 is formed in a press to provide a layer that exhibits a desired thickness, area and density.

In some embodiments, the average length of the CNTs 112 forming the matrix 110 may be at least 0.1 μm , 0.5 μm , 1 μm , 5 μm , 10 μm , 50 μm , 100 μm , 200 μm , 300 μm , 400

μm , 500 μm , 600 μm , 7000 μm , 800 μm , 900 μm , 1,000 μm or more. For example, in some embodiments, the average length of the CNTs 112 forming the matrix 110 may be in the range of 1 μm to 1,000 μm , or any subrange thereof, such as 1 μm to 600 μm . In some embodiments, more than 50%, 60%, 70%, 80%, 90%, 95%, 99% or more of the CNTs 112 may have a length within 10% of the average length of the CNTs 112 making up the matrix 110.

In various embodiments, the other carbonaceous material 108 can include carbon in a variety of forms, including activated carbon, carbon black, graphite, and others. The carbonaceous material can include carbon particles, including nanoparticles, such as nanotubes, nanorods, graphene in sheet, flake, or curved flake form, and/or formed into cones, rods, spheres (buckyballs) and the like.

Applicants have found unexpected result that an active layer of the type herein can provide exemplary performance (e.g., high conductivity, low resistance, high voltage performance, and high energy and power density) even when the mass fraction of CNTs in the layer is quite low. For example, in some embodiments, the active layer may be at least about 50 wt %, 60 wt %, 70 wt %, 75 wt %, 80 wt %, 85 wt %, 90 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt %, 99.5 wt %, or more elemental carbon in a form other than CNT (e.g., activated carbon). In particular, for certain applications involving high performance ultracapacitors, active layers 106 that are in the range of 95 wt % to 99 wt % activated carbon (with the balance CNTs 112), have been shown to exhibit excellent performance.

In some embodiments, the matrix 110 of CNTs 112 form an interconnected network of highly electrically conductive paths for current flow (e.g. ion transport) through the active layer 106. For example, in some embodiments, highly conductive junctions may occur at points where CNTs 112 of the matrix 110 intersect with each other, or where they are in close enough proximity to allow for quantum tunneling of charge carriers (e.g., ions) from one CNT to the next. While the CNTs 112 may make up a relatively low mass fraction of the active layer (e.g., less than 10 wt %, 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt % or less, e.g., in the range of 0.5 wt % to 10 wt % or any subrange thereof such as 1 wt % to 5.0 wt %), the interconnected network of highly electrically conductive paths formed in the matrix 110 may provide long conductive paths to facilitate current flow within and through the active layer 106 (e.g. conductive paths on the order of the thickness of the active layer 106).

For example, in some embodiments, the matrix 110 may include one or more structures of interconnected CNTs, where the structure has an overall length in along one or more dimensions longer than 2, 3, 4, 5, 10, 20, 50, 100, 500, 1,000, 10,000 or more times the average length of the component CNTs making up the structure. For example, in some embodiments, the matrix 110 may include one or more structures of interconnected CNTs, where the structure has an overall in the range of 2 to 10,000 (or any subrange thereof) times the average length of the component CNTs making up the structure. For example, in some embodiments the matrix 110 may include highly conductive pathways having a length greater than 100 μm , 500 μm , 1,000 μm , 10,000 μm or more, e.g., in the range of 100 μm -10,000 μm of any subrange thereof.

As used herein, the term “highly conductive pathway” is to be understood as a pathway formed by interconnected CNTs having an electrical conductivity higher than the electrical conductivity of the other carbonaceous material 108 (e.g., activated carbon), surrounding that matrix 110 of CNTs 112.

Not wishing to be bound by theory, in some embodiments the matrix **110** can be characterized as an electrically interconnected network of CNT exhibiting connectivity above a percolation threshold. Percolation threshold is a mathematical concept related to percolation theory, which is the formation of long-range connectivity in random systems. Below the threshold a so called “giant” connected component of the order of system size does not exist; while above it, there exists a giant component of the order of system size.

In some embodiments, the percolation threshold can be determined by increasing the mass fraction of CNTs **112** in the active layer **106** while measuring the conductivity of the layer, holding all other properties of the layer constant. In some such cases, the threshold can be identified with the mass fraction at which the conductivity of the layer sharply increases and/or the mass fraction above which the conductivity of the layer increases only slowly with increases with the addition of more CNTs. Such behavior is indicative of crossing the threshold required for the formation of interconnected CNT structures that provide conductive pathways with a length on the order of the size of the active layer **106**.

Returning to FIG. 1, in some embodiments, one or both of the active layer **106** and the adhesion layer **104** may be treated by applying heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like). For example, in some embodiments, one or both of the layers can be heated to at least 100 C, 150 C, 200 C, 250 C, 300 C, 350 C, 400 C, 450 C, 500 C or more for at least 1 minute, 5 minutes, 10 minutes, 30 minutes, 1 hour, 2 hours, 3 hours, 12 hours, 24 hours, or more. For example, in some embodiments the layers may be treated to reduce moisture in the layer to less than 1,000 ppm, 500 ppm, 100 ppm, 10 ppm, 1 ppm, 0.1 ppm or less.

Returning to FIG. 1, in some embodiments, the adhesion layer **104** may be formed of carbon nanotubes. For example, in some embodiments, the adhesion layer **104** may be at least about 50%, 75%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, 99.999% by mass CNTs. In some embodiments, the CNTs may be grown directly on the conductive layer **102**, e.g., using the chemical vapor deposition techniques such as those described in U.S. Patent Pub. No. 2013/0045157 entitled “In-line Manufacture of Carbon Nanotubes” and published Feb. 21, 2013. In some embodiments, the CNTs may be transferred onto the conductive layer **102**, e.g., using wet or dry transfer processes, e.g., of the type described e.g., in U.S. Patent Pub. No. 2013/0044405 entitled “High Power and High Energy Electrodes Using Carbon Nanotubes” and published Feb. 21, 2013. In some embodiments, the adhesion layer **104** adheres to the overlying active layer **106** using substantially only electrostatic forces (e.g., Van Der Waals attractions) between the CNTs of the adhesion layer **104** and the carbon material and CNTs of the active layer **106**.

In some embodiments, the CNTs of the adhesion layer **104** may include single wall nanotubes (SWNT), double wall nanotubes (DWNT), or multiwall nanotubes (MWNT), or mixtures thereof. In some embodiments the CNTs may be vertically aligned. In one particular embodiment, the CNTs of the adhesion layer **104** may be primarily or entirely SWNTs and/or DWNTs, while the CNTs of the active layer **106** a primarily or entirely MWNTs. For example, in some embodiments, the CNTs of the of the adhesion layer **104** may be at least 75%, at least 90%, at least 95%, at least 99% or more SWNT or at least 75%, at least 90%, at least 95%, at least 99% or more DWNT. In some embodiments, the CNTs of the of the active layer **106** may be at least 75%, at least 90%, at least 95%, at least 99% or more MWNT.

In some embodiments, the adhesion layer **104** may be formed by applying pressure to a layer of carbonaceous material. In some embodiments, this compression process alters the structure of the adhesion layer **104** in a way that promotes adhesion to the active layer **106**. For example, in some embodiments pressure may be applied to layer comprising a vertically aligned array of CNT or aggregates of vertically aligned CNT, thereby deforming or breaking the CNTs.

In some embodiments, the adhesion layer may be formed by casting a wet slurry of CNTs (with or without additional carbons) mixed with a solvent onto the conductive layer **102**. In various embodiments, similar techniques to those described above for the formation of the active layer **106** from a wet slurry may be used.

In some embodiments, mechanical energy may be introduced to the wet slurry using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

In some embodiments, the solid carbon fraction of the wet slurry may be less than 10 wt %, 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt %, 0.5 wt %, 0.1 wt % or less, e.g., in the range of 0.1 wt % to 10 wt % or any subrange thereof such as 0.1 wt % to 2 wt %.

In various embodiments, the conductive layer **102** may be made of a suitable electrically conductive material such as a metal foil (e.g., an aluminum foil). In some embodiments, the surface of the conductive layer **102** may be roughened, patterned, or otherwise texturized, e.g., to promote adhesion to the adhesion layer **104** and good electrical conductance from the active layer **106**. For example, in some embodiments, the conductive layer may be etched (e.g., mechanically or chemically). In some embodiments, the conductive layer **102** may have a thickness in the range of 1 μ m to 1,000 μ m or any subrange thereof such as 5 μ m to 50 μ m.

In some embodiments, the conductive layer **102** may include a nanostructured surface. For example, as described in International Pub. No. WO 2016/057983 entitled “Nanostructured Electrode for Energy Storage Device” published Apr. 14, 2016, the conductive layer may have a top surface that includes nanoscale features such as whiskers (e.g., carbide whiskers) that promote adhesion to the adhesion layer **104** and good electrical conductance from the active layer **106**. An exemplary current collector is the current collector available from Toyo Aluminum K.K. under the trade name TOYAL-CARBO®.

In some embodiments, one or both of the active layer **106** and the adhesion layer **104** may be treated by applying heat and/or vacuum to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

In some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be compressed, e.g., to break some of the constituent CNTs or other carbonaceous material to increase the surface area of the respective layer. In some embodiments, this compression treatment may increase one or more of adhesion between the layers, ion transport rate within the layers, and the surface area of

the layers. In various embodiments, compression can be applied before or after the respective layer is applied to or formed on the electrode **100**.

In some embodiments, the adhesion layer **104** may be omitted, such that the active layer **106** is disposed directly on the conductive layer **102**.

Referring to FIG. 3, in some embodiments, the electrode **100** may be double sided, with an adhesion layer **104** and active layer **106** formed on each of two opposing major surfaces of the conductive layer **102**. In some embodiments, the adhesion layer **104** may be omitted on one or both sides of the two-sided electrode **100**.

Referring to FIG. 4, an exemplary embodiment of method **200** of making the active layer **106** of electrode **100** is described. In step **201**, CNTs are dispersed in a solvent to form a dispersion of CNTs. In some embodiments, the dispersion may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405 entitled "High Power and High Energy Electrodes Using Carbon Nanotubes" published Feb. 21, 2013 including stirring, sonication, or a combination of the two. In various embodiments, any suitable solvent may be used, including, for example, ethanol, methanol, isopropyl alcohol, dimethyl sulfoxide, dimethylformamide, acetone, acetonitrile, and the like. In general, it is advantageous to choose a solvent that will be substantially eliminated in the drying step **204** described below, e.g., using heat and/or vacuum drying techniques.

In some embodiments, the mixture of CNTs and solvents may be passed through a filter, e.g., an array of micro channels (e.g., having channels with diameters on the order of the radial size of the CNTs) to help physically separate the CNTs and promote dispersion.

In some embodiments, the CNT dispersion may be formed without the addition of surfactants, e.g., to avoid the presence of impurities derived from these surfactants at the completion of the method **200**.

In step **202**, the CNT dispersion is mixed with carbonaceous material (e.g., activated carbon) to form a slurry. In some embodiments, the slurry may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405, including stirring, sonication, or a combination of the two. In some embodiments, the slurry may have solid carbon fraction of less than 20 wt %, 15 wt %, 10 wt %, 5 wt %, 2 wt %, 1 wt %, or less, e.g., in the range of 1 wt % to 20 wt % or any subrange thereof such as 4% to 6%. The mass ratio of CNTs to other carbonaceous material in the slurry may be less than 1:5, 1:10, 1:15, 1:20, 1:50, 1:100, or less, e.g., in the range of 1:10 to 1:20 or any subrange thereof.

In step **203**, the slurry is applied to either the adhesion layer **104** or, if the adhesion layer **104** is omitted, the conductive layer **102** of the electrode **100**. In some embodiments, the slurry may be formed into a sheet, and coated onto the electrode. For example, in some embodiments, slurry may be applied to through a slot die to control the thickness of the applied layer. In other embodiments, the slurry may be applied to the conductive layer **102**, and then leveled to a desired thickness, e.g., using a doctor blade.

In some embodiments, the slurry may be compressed (e.g., using a calendaring apparatus) before or after being applied to the electrode **100**. In some embodiments, the slurry may be partially or completely dried (e.g., by applying heat, vacuum or a combination thereof) during this step **203**.

In step **204**, if the slurry has not dried, or has been only partially dried during step **203**, the slurry applied to the electrode is fully dried, (e.g., by applying heat, vacuum or a

combination thereof). In some embodiments, substantially all of the solvent (and any other non-carbonaceous material such as dispersing agents) is removed from the active layer **106**. In some embodiments, if impurities remain following the drying step, and additional step of heating (e.g. baking or annealing) the layer may be performed. For example, in some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be treated by applying heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

Referring to FIG. 5, an exemplary embodiment of method **300** of making the adhesion layer **104** of electrode **100** is described. In step **301**, CNTs are dispersed in a solvent to form a dispersion of CNTs. In some embodiments, the dispersion may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405 including stirring, sonication, or a combination of the two. In various embodiments, any suitable solvent may be used, including, for example an organic solvent such as isopropyl alcohol, acetonitrile or propylene carbonate. In general, it is advantageous to choose a solvent that will be substantially eliminated in the drying step **304** described below.

In some embodiments, the mixture of CNTs and solvents may be passed through a filter, e.g., an array of micro channels (e.g., having channels with diameters on the order of the radial size of the CNTs) to help physically separate the CNTs and promote dispersion.

In some embodiments, the CNT dispersion may be formed without the addition of surfactants, e.g., to avoid the presence of impurities derived from these surfactants at the completion of the method **300**.

In step **302**, the CNT dispersion may optionally be mixed with additional carbonaceous material (e.g., activated carbon) to form a slurry. In some embodiments, the additional carbonaceous material may be omitted, such that the slurry is made up of CNTs dispersed in a solvent. In some embodiments, the slurry may have solid fraction of less than 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt %, 0.5 wt %, 0.1 wt %, or less, e.g., in the range of 0.1 to 5 wt % or any subrange thereof.

In step **303**, the slurry is applied to the conductive layer **102** of the electrode **100**. In some embodiments, the slurry may be coated onto the electrode. For example, in some embodiments, slurry may be applied to through a slot die to control the thickness of the applied layer. In other embodiments, the slurry may be applied to the conductive layer **102**, and then leveled to a desired thickness, e.g., using a doctor blade.

In some embodiments, the slurry may be compressed (e.g., using a calendaring apparatus) before or after being applied to the electrode **100**. In some embodiments, the slurry may be partially or completely dried (e.g., by applying heat, vacuum or a combination thereof) during this step **303**.

In step **304**, if the slurry has not dried, or has been only partially dried during step **203**, the slurry applied to the electrode is fully dried, (e.g., by applying heat, vacuum or a combination thereof). In some embodiments, substantially all of the solvent (and any other non-carbonaceous material such as dispersing agents) is removed from the active layer **106**. In some embodiments, if impurities remain following the drying step, and additional step of heating (e.g. baking or annealing) the layer may be performed. For example, in some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be treated by applying

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heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

In some embodiments, the method 300 for forming an adhesion layer 104 and method 200 for forming an active layer 106 may be performed in series to successively form the adhesion layer 104 followed by the overlaying active layer 106. In some embodiments, the foregoing methods may be repeated, e.g., to form a two-sided electrode of the type described herein.

Advantageously, in some embodiments, the method 300 for forming an adhesion layer 104 and/or method 200 for forming an active layer 106 may be implemented as a roll-to-roll processes, e.g., to allow volume production of electrode sheets several tens of meters long or more.

FIG. 6 shows an exemplary mixing apparatus 400 for implementing the method 300 for forming an adhesion layer 104 and/or method 200 for forming an active layer 106. In the interest of brevity, the apparatus 400 will be described for use in forming active layer 106 using method 200. However, as will be apparent to one skilled in the art, the apparatus 400 can easily be configured to implement the method 300 for forming an adhesion layer 104.

The apparatus 400 includes a mixing vessel 401. The mixing vessel receives a slurry composed of a solvent, carbon nanotubes, and (optionally) additional carbonaceous material of the type described above. In some embodiments, this slurry (or components thereof) may be initially formed in the mixing vessel 401. In other embodiments, the slurry may be formed elsewhere and then transferred to the mixing vessel 401.

In some embodiments the mixing vessel 401 may include one or more mechanisms for mixing the slurry, such as an impeller or high shear mixer. In some embodiments, a mixing mechanism may be provided which is capable of stirring the slurry at a controlled rate, e.g., of up to 1000 rotations per minute (RPM) or more. In some embodiments, the mixing vessel may include one or more devices for applying mechanical energy to the slurry, such as a sonicator, mixer (e.g., a high shear mixer), homogenizer, or any other suitable device known in the art. In some embodiments, the mixing vessel may be temperature controlled, e.g., using one or more heating and/or cooling elements such as electric heaters, tubing for circulating chilled water, or any other such devices known in the art.

Slurry from the mixing vessel 401 may be circulated through a flow line 402, e.g., a pipe or tubing, using a pump 403. Pump 403 may be any suitable configuration, such as a peristaltic pump. A flow meter 404 may be provided to measure the rate of slurry flow through the flow line 402. A filter 405 may be provided to filter the slurry flowing through the flow line 402, e.g., to remove clumps of solid material having a size above a desired threshold.

In some embodiments, e.g., where mixing vessel 401 does not include a sonicator, an in-line sonicator 406 may be provided to sonicate slurry flowing through the flow line 402. For example, in some embodiments a flow through sonicator such as the Branson Digital SFX-450 sonicator available commercially from Thomas Scientific of 1654 High Hill Road Swedesboro, N.J. 08085 U.S.A may be used.

In some embodiments, a temperature control device 407, such as a heat exchanger arranged in a sleeve disposed about the flow line 402, is provided to control the temperature of the slurry flowing through the flow line 402.

In some embodiments a valve 408 is provided which can be selectively controlled to direct a first portion of the slurry flowing through flow line 402 to be recirculated back to the

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mixing vessel 401, while a second portion is output externally, e.g., to a coating apparatus 500.

In some embodiments, a sensor 409 such as a pressure sensor or flow rate sensor is provided to sense one or more aspects of the output portion of slurry.

In various embodiments any or all of the elements of apparatus 400 may be operatively connected to one or more computing devices to provide for automatic monitoring and/or control of the mixing apparatus 400. For example, the sonicator 406 may include digital controls for controlling its operating parameters such as power and duty cycle.

In various embodiments, the coating apparatus 500 may be any suitable type known in the art. For example, FIG. 7A shows an exemplary embodiment of coating apparatus 500 featuring a slot die 501 that distributes slurry received from a source such as the mixing apparatus 400 through a distribution channel 502 onto a substrate 503 (e.g., the conductive layer 102, either bare or already coated with adhesion layer 104) which moves across a roller 504. Setting the height of the slot die above the substrate 503 on the roller 504 and controlling the flow rate and/or pressure of the slurry in the channel 502 allows for control of the thickness and density of the applied coating. In some embodiments, channel 502 may include one or more reservoirs to help ensure consistent flow of slurry to provide uniform coating during operation.

FIG. 7B shows an exemplary embodiment of coating apparatus 500 featuring a doctor blade 601 that levels slurry received from a source such as the mixing apparatus 400 that is applied through one or more applicators 602 (one is shown) onto a substrate 603 (e.g., the conductive layer 102, either bare or already coated with adhesion layer 104) which moves across a roller 604. The direction of travel of the substrate 603 is indicated by the heavy dark arrow. Setting the height of the doctor blade 601 above the substrate 603 on the roller 604 and controlling the flow rate and/or pressure of the slurry through the applicator 602 allows for control of the thickness and density of the applied coating. Although a single doctor blade 601 is shown, multiple blades may be used, e.g., a first blade to set a rough thickness of the coating, and a second blade positioned down line from the first blade to provide fine smoothing of the coating.

Further, disclosed herein are capacitors incorporating the electrode that provide users with improved performance in a wide range of temperatures. Such ultracapacitors may comprise an energy storage cell and an electrolyte system within an hermetically sealed housing, the cell electrically coupled to a positive contact and a negative contact, wherein the ultracapacitor is configured to operate at a temperatures within a temperature range between about -100 degrees Celsius to about 300 degrees Celsius or more, or any subrange thereof, e.g., -40 C to 200 C, -40 C to 250 C, -40 C to 300 C, 0 C to 200 C, 0 C to 250 C, 0 C to 300 C. In some embodiments such ultracapacitors can operate at voltages of 1.0 V, 2.0 V, 3.0 V, 3.2 V, 3.5 V, 4.0 V, or more, e.g., for lifetimes exceeding 1,000 hours.

As shown in FIGS. 8A and 8B, exemplary embodiments of a capacitor are shown. In each case, the capacitor is an "ultracapacitor 10." The difference between FIG. 8A and FIG. 8B is the inclusion of a separator in exemplary ultracapacitor 10 of FIG. 8A. The concepts disclosed herein generally apply equally to any exemplary ultracapacitor 10. Certain electrolytes of certain embodiments are uniquely suited to constructing an exemplary ultracapacitor 10 without a separator. Unless otherwise noted, the discussion herein applies equally to any ultracapacitor 10, with or without a separator.

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The exemplary ultracapacitor **10** is an electric double-layer capacitor (EDLC). The EDLC includes at least one pair of electrodes **3** (where the electrodes **3** may be referred to as a negative electrode **3** and a positive electrode **3**, merely for purposes of referencing herein). When assembled into the ultracapacitor **10**, each of the electrodes **3** (which may each be an electrode **100** of the type shown in FIG. 1 above) presents a double layer of charge at an electrolyte interface. In some embodiments, a plurality of electrodes **3** is included (for example, in some embodiments, at least two pairs of electrodes **3** are included). However, for purposes of discussion, only one pair of electrodes **3** are shown. As a matter of convention herein, at least one of the electrodes **3** uses a carbon-based energy storage media **1** (e.g., the active layer **106** of electrode **100** shown in FIG. 1), and it is assumed that each of the electrodes includes the carbon-based energy storage media **1**. It should be noted that an electrolytic capacitor differs from an ultracapacitor because metallic electrodes differ greatly (at least an order of magnitude) in surface area.

Each of the electrodes **3** includes a respective current collector **2** (also referred to as a “charge collector”), which may be the conductive layer **102** of electrode **100** shown in FIG. 1. In some embodiments, the electrodes **3** are separated by a separator **5**. In general, the separator **5** is a thin structural material (usually a sheet) used to separate the negative electrode **3** from the positive electrode **3**. The separator **5** may also serve to separate pairs of the electrodes **3**. Once assembled, the electrodes **3** and the separator **5** provide a storage cell **12**. Note that, in some embodiments, the carbon-based energy storage media **1** may not be included on one or both of the electrodes **3**. That is, in some embodiments, a respective electrode **3** might consist of only the current collector **2**. The material used to provide the current collector **2** could be roughened, anodized or the like to increase a surface area thereof. In these embodiments, the current collector **2** alone may serve as the electrode **3**. With this in mind, however, as used herein, the term “electrode **3**” generally refers to a combination of the energy storage media **1** and the current collector **2** (but this is not limiting, for at least the foregoing reason).

At least one form of electrolyte **6** is included in the ultracapacitor **10**. The electrolyte **6** fills void spaces in and between the electrodes **3** and the separator **5**. In general, the electrolyte **6** is a substance that dissociates into electrically charged ions. A solvent that dissolves the substance may be included in some embodiments of the electrolyte **6**, as appropriate. The electrolyte **6** conducts electricity by ionic transport.

In some embodiments, the electrolyte **6** may be in gelled or solid form (e.g., an ionic liquid impregnated polymer layer). Examples of such electrolytes are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

In other embodiments, the electrolyte **6** may be in non-aqueous liquid form, e.g., an ionic liquid, e.g., of a type suitable for high temperature applications. Examples of such electrolytes are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

In some embodiments, the storage cell **12** is formed into one of a wound form or prismatic form which is then packaged into a cylindrical or prismatic housing **7**. Once the electrolyte **6** has been included, the housing **7** may be

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hermetically sealed. In various examples, the package is hermetically sealed by techniques making use of laser, ultrasonic, and/or welding technologies. In addition to providing robust physical protection of the storage cell **12**, the housing **7** is configured with external contacts to provide electrical communication with respective terminals **8** within the housing **7**. Each of the terminals **8**, in turn, provides electrical access to energy stored in the energy storage media **1**, generally through electrical leads which are coupled to the energy storage media **1**.

As discussed herein, “hermetic” refers to a seal whose quality (i.e., leak rate) is defined in units of “atm-cc/second,” which means one cubic centimeter of gas (e.g., He) per second at ambient atmospheric pressure and temperature. This is equivalent to an expression in units of “standard He-cc/sec.” Further, it is recognized that 1 atm-cc/sec is equal to 1.01325 mbar-liter/sec. Generally, the ultracapacitor **10** disclosed herein is capable of providing a hermetic seal that has a leak rate no greater than about 5.0×10^{-6} atm-cc/sec, and may exhibit a leak rate no higher than about 5.0×10^{-10} atm-cc/sec. It is also considered that performance of a successfully hermetic seal is to be judged by the user, designer or manufacturer as appropriate, and that “hermetic” ultimately implies a standard that is to be defined by a user, designer, manufacturer or other interested party.

Leak detection may be accomplished, for example, by use of a tracer gas. Using tracer gas such as helium for leak testing is advantageous as it is a dry, fast, accurate and non destructive method. In one example of this technique, the ultracapacitor **10** is placed into an environment of helium. The ultracapacitor **10** is subjected to pressurized helium. The ultracapacitor **10** is then placed into a vacuum chamber that is connected to a detector capable of monitoring helium presence (such as an atomic absorption unit). With knowledge of pressurization time, pressure and internal volume, the leak rate of the ultracapacitor **10** may be determined.

In some embodiments, at least one lead (which may also be referred to herein as a “tab”) is electrically coupled to a respective one of the current collectors **2**. A plurality of the leads (accordingly to a polarity of the ultracapacitor **10**) may be grouped together and coupled to into a respective terminal **8**. In turn, the terminal **8** may be coupled to an electrical access, referred to as a “contact” (e.g., one of the housing **7** and an external electrode (also referred to herein for convention as a “feed-through” or “pin”). Suitable exemplary designs are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

Various forms of the ultracapacitor **10** may be joined together. The various forms may be joined using known techniques, such as welding contacts together, by use of at least one mechanical connector, by placing contacts in electrical contact with each other and the like. A plurality of the ultracapacitors **10** may be electrically connected in at least one of a parallel and a series fashion.

As used herein the symbol “wt %” means weight percent. For example, when referring to the weight percent of a solute in a solvent, “wt %” refers to the percentage of the overall mass of the solute and solvent mixture made up by the solute.

The entire contents of each of the publications and patent applications mentioned above are incorporated herein by reference. In the event that the any of the cited documents conflicts with the present disclosure, the present disclosure shall control.

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While the invention has been described with reference to exemplary embodiments, it will be understood that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. For example, in some embodiments, one of the foregoing layers may include a plurality of layers there within. In addition, many modifications will be appreciated to adapt a particular instrument, situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims.

What is claimed is:

1. An electrode for an energy storage device, the electrode comprising:

a conductive layer;

an active layer of energy storage media substantially free from binding agents and comprising a matrix of carbon nanotubes defining void spaces and binding carbonaceous material in the void spaces; and

an adhesion layer being substantially free of binder material and disposed between the active layer and the conductive layer;

wherein the matrix of carbon nanotubes comprises a concentration of carbon nanotubes that exceeds a percolation threshold thus causing the electrode to exhibit electrical connectivity within the active layer;

wherein the matrix of carbon nanotubes is less than 1% by weight of the active layer; and wherein the carbon nanotubes comprise aligned carbon nanotubes.

2. The electrode as in claim 1, wherein the matrix of carbon nanotubes comprises conductive paths configured to facilitate current flow within and through the active layer.

3. The electrode as in claim 1, wherein the carbon nanotubes (CNT) comprise at least one of single wall nanotubes (SWNT), double wall nanotubes (DWNT), multiwall nanotubes (MWNT) and vertically aligned nanotubes.

4. The electrode as in claim 1, wherein the conductive layer comprises a conductive material, and wherein the conductive material comprises a metal foil.

5. The electrode as in claim 1, wherein a surface of the conductive layer is at least one of patterned, roughened and textured.

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6. The electrode as in claim 1, wherein a surface of the conductive layer comprises a nanostructured surface that promotes adhesion thereto.

7. An energy storage device comprising the electrode as in claim 1, wherein the energy storage device is one of an ultracapacitor and a battery.

8. The electrode as in claim 1, wherein the active layer comprises at least ninety (90) weight percent carbonaceous material.

9. The electrode as in claim 1, wherein at least one of the active layer and the adhesion layer is a compressed layer substantially free from impurities.

10. An energy storage device comprising:

an electrode comprising a conductive layer, an adhesion layer disposed on the conductive layer, and an active layer disposed on the adhesion layer;

the conductive layer comprising a metal foil that comprises at least one of a roughened, patterned, textured and nanostructured surface for promoting adhesion of the adhesion layer;

the adhesion layer being substantially free of binder material and comprising carbonaceous material and an adhesion layer matrix of at least fifty percent aligned carbon nanotubes by weight of the adhesion layer, the carbonaceous material being bound in void spaces defined by the adhesion layer matrix; and

the active layer being substantially free of binder material and comprising carbonaceous material and an active layer matrix of less than one percent aligned carbon nanotubes by weight of the active layer, the carbonaceous material being bound in void spaces defined by the active layer matrix, the active layer matrix comprising a concentration of carbon nanotubes that exceeds a percolation threshold thus causing the electrode to exhibit electrical connectivity within the active layer;

wherein the active layer is thicker than the adhesion layer.

11. The energy storage device as in claim 10, which is one of a battery and an ultracapacitor.

12. The energy storage device as in claim 10, wherein at least one of the active layer and the adhesion layer is a compressed layer substantially free from impurities.

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